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6MMCS009W1 Usability Testing & Evaluation

Group Coursework - Experimental Operandum

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Experimental Operandum

1 Group information

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2 Introduction and Objectives

This research aims to evaluate the usability of the Sri Lankan Government Website Parliament.lk. This research will assess key usability metrics such as Efficacy, Memorability, Learnability, and Error Rate of the aforementioned site. The findings from this analysis will allow developers to make more informed decisions regarding future website modifications.

3 Research Questions and Hypotheses

Name	Usability Metrics	Statistical Hypotheses test
Thananchjayan Mahendran	Learnability	Pearson correlation
Hansaka Tennakoon	Efficiency	Paired-sample t-test
Muhammed Faizal	Satisfaction	Independent-sample t-test
Dulan Sanjula	Error Rate	ANOVA

3.1 Independent-sample-test

Student Name: Muhammed Faizal

Research question	How satisfied are you with the ease of finding the Hansard report on parliament.lk	
Usability metric studied	Satisfaction – Evaluating how satisfied users are with the website layout after completing a specific task.	
Hypothesis (statement)	**Null Hypothesis (H_0):** There is no significant difference in satisfaction levels between users of different age groups (18-30 and above 30) in locating the Hansard report on parliament.lk.	
	Alternative Hypothesis (H_1):There is a significant difference in satisfaction levels between users of different age groups (18-30 and above 30) in locating the Hansard report on parliament.lk.	
Mathematical Expression	**Null Hypothesis (H_0):** $\rho = 0$ The mean satisfaction score of group 1 (18-30) is equal to the mean satisfaction score of group 2 (above 30).	
	Alternative Hypothesis (H_1): $\rho \neq 0$ The mean satisfaction score of group 1 (18-30) is not equal to the mean satisfaction score of group 2 (above 30)	
Variables	Independent	Age group (18–30 vs. above 30).
	Dependent	Satisfaction score measured on a Likert scale, from 1–5 on how easy it was to find the Hansard report
	Control	Task given (the same task of locating the Hansard report on parliament.lk).
	Measurement Scale	Likert Scale: Participants rate their satisfaction on a scale of 1 to 5, where: 1 = Very Dissatisfied 2 = Dissatisfied 3 = Neutral 4 = Satisfied 5 = Very Satisfied
	Data type	**Quantitative Data:** Satisfaction scores are numerical (1–5 scale). **Ordinal Data:** Since the Likert scale measures satisfaction, it is ordered data. **Nominal Data:** User age is categorical
Experimental design	**Between-Subjects Design:** The participants are divided into two separate groups based on their age: 18-30 and above 30. Each group will perform the same task, but no one will be in both groups.	
Tasks and Scenarios	**Task:** Participants will be asked to locate the Hansard report on parliament.lk. **Scenario:** You are looking for a specific Hansard report on the parliament.lk website. Please rate how easy or difficult it was to find the report.	

Procedure	The data is collected remotely through an online survey after the users complete the task. The task will be unmoderated, allowing participants to work on it in their own time to mimic a natural experience.
Data collection method	The data is collected using an online survey tool (Google Forms), where participants submit their satisfaction ratings.
Test material	Questionnaire for Satisfaction Rating: Sample Question: How satisfied were you with the overall ease of use of parliament.lk during the task? (Rate from 1–5). Instructions to Participants: Read task instructions and complete the task within the allotted time. Record your satisfaction immediately after finishing the task. Measurement Details: **Start of Task:** When the participant begins searching for the report. **End of Task:** When the participant found the report.

3.2 Paired-sample-test

Student Name: Hansaka Tennakoon

Research question	How does task completion time differ when using a guided workflow compared to an unguided workflow within Application ?	
Usability metric studied	**Efficiency**: Measuring how quickly users can complete tasks when provided with a guided workflow versus no guidance.	
Hypothesis (statement)	**Null Hypothesis (H_0):** There is no significant difference in task completion time between the guided and unguided workflow.	
	Alternative Hypothesis (H_1): There is a significant difference in task completion time between the guided and unguided workflow.	
Mathematical Expression	**Null Hypothesis (H_0):** $\mu_1 = \mu_2$ (The mean task completion time for the guided workflow is equal to the mean task completion time for the unguided workflow.)	
	Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$ (The mean task completion time for the guided workflow is not equal to the mean task completion time for the unguided workflow.)	
Variables	Independent	Workflow guidance(guided or unguided)
	Dependent	Task completion time
	Control	The specific task assigned to participants.
	Measurement Scale	Task completion time is measured in seconds
	Data type	Continuous data (task completion time).
Experimental design	**Within-subjects design**: ● Reason: Each participant completes the same task under both conditions (guided and unguided workflow). This ensures that differences in task completion time are due to the workflow condition rather than individual participant variability.	
Tasks and Scenarios	**Participants are asked to complete a specific task twice**: 1. Guided Workflow Condition: Participants complete the task with step-by-step instructions, tooltips, or hints provided. 2. Unguided Workflow Condition: Participants complete the same task without guidance or assistance.	
Procedure	● Type: Moderated usability testing conducted remotely. ● Justification: Remote testing allows participants to join from their preferred locations, increasing convenience and participation rates, while moderation ensures consistent instructions and troubleshooting for any technical issues.	
Data collection method	**Tools Used**: ● Screen-recording software to capture task performance and completion times. ● System log files for additional timestamped data. **Justification**: Screen recordings provide a visual and time-stamped record of user interactions, ensuring accurate measurement of task start and end times. System logs act as a backup and validation for the recorded data.	

Test material	**Task Details**: Participants are provided with a task description to complete under two conditions (guided and unguided). ● For the guided workflow, detailed step-by-step instructions or contextual tooltips are displayed. ● For the unguided workflow, participants are only given the task description without further guidance. **Instructions**: Participants are instructed to complete the task as quickly and accurately as possible. **Measurement**: ● Task start: The moment the participant begins the task ● Task end: The moment the task is completed ● Task completion time is recorded in seconds.

3.3 ANOVA

Student Name: Dulan Sanjula

Research question	How does the layout of the parliament.lk website on different devices affect the usability.	
Usability metric studied	**Error Rate :** How long it will take users to find and download the hansard document for the stated date between two different devices.	
Hypothesis (statement)	Null Hypothesis (H_0): There is no significant difference in the time taken for the user to retrieve the hansard report between the two device types.	
	Alternative Hypothesis (H_1): There is a significant difference in the time taken for the user to retrieve the hansard report between the two device types.	
Mathematical Expression	Null Hypothesis (H_0): $\mu1=\mu2$	
	Alternative Hypothesis (H_1): $\mu1\neq\mu2$	
Variables	Independent	The device used to complete the task (Desktop/Laptop vs Smartphone)
	Dependent	The Time taken to complete the task.
	Control	Task 1 : Navigate to the Hansard Report for the given date on a Laptop or Desktop Computer. Task 2: Navigate to the Hansard Report for the given date on a Smartphone Device.
	Measurement Scale	Time in minutes and seconds form completion
	Data type	Continuous Data
Experimental design	Within-subject design was chosen since the same participant will be able to complete both tasks, therefore lowering the chance of external factors such as Internet Speed or Measurement error from affecting the results.	
Tasks and Scenarios	1. Laptop/Desktop Workflow Condition : The participant is instructed to navigate to and download the Hansard Report for the given date on a Laptop or Desktop. 2. Smartphone Condition : The participant is instructed to navigate to and download the Hansard Report for the given date on a Smartphone.	
Procedure	● Type: Moderated usability testing conducted remotely. ● Justification: Remote testing allows participants to join from their preferred locations, increasing convenience and participation rates, while moderation ensures consistent instructions and troubleshooting for any technical issues.	
Data collection method	The data is collected using an online survey tool where participants are guided through the study; this allows participants to complete the survey at their own convenience.	
Test material	The data collected from Stage 4 and the Mobile stages of the questionnaire are considered for this hypothesis. The Participant is instructed to locate hansard report for the 18th July 2023 on both laptop/desktop and Smartphone devices and to record the time in minutes and seconds it takes them to complete the task on both device types.	

	Measurement: ● Task start: The moment the participant begins the task ● Task end: The moment the task is completed ● Task completion time is recorded in minutes and seconds.

3.4 Pearson Correlation

Student Name: Thananchjayan Mahendran

Research question	How effectively can first-time users learn to navigate and perform essential tasks on the Parliament.lk government website?	
Usability metric studied	**Learnability:** How easily new users can learn to use the system and become proficient in its functionality with minimal effort and time.	
Hypothesis (statement)	**Null Hypothesis (H_0):** There is no significant correlation between the usability factors of the Parliament.lk government website and the effectiveness of first-time users in learning to navigate and perform essential tasks.	
	Alternative Hypothesis (H_1): There is a significant correlation between the usability factors of the Parliament.lk government website and the effectiveness of first-time users in learning to navigate and perform essential tasks.	
Mathematical Expression	**Null Hypothesis (H_0): $\rho = 0$** Where ρ (rho) represents the population correlation coefficient between usability factors and the effectiveness of first-time users.	
	Alternative Hypothesis (H_1): $\rho \neq 0$ This indicates that there is a significant correlation (positive or negative) between usability factors and the effectiveness of first-time users.	
Variables	Independent	navigation design
	Dependent	post-task confidence ratings
	Control	The specific task given to the participants
	Measurement Scale	Users' ratings after completing the task by themselves.
	Data type	Ratings, typically collected via Likert scales, fall under the ordinal data type. (Ordinal)
Experimental design	**Within-Subjects Design,** ● Consistency Across Participants: All participants are exposed to the same usability tasks, allowing for a direct comparison of learnability metrics ● Focus on Learnability: Learnability requires assessing how quickly and effectively a single user can adapt to performing tasks on the website. Measuring this across multiple tasks for the same participants helps establish a pattern.	
Tasks and Scenarios	**Participants asked to complete the tasks three times.** 1. First the participants asked to do the specific task? 2. Then we guided step by step to do the task by providing a screenshot of every step. 3. Once again we asked them to do the task by themselves without any guidance.	
Procedure	● **Type:** Moderated usability testing conducted remotely. ● **Justification:** Moderation guarantees consistent instructions and support for any technological challenges, while remote testing enables participants to participate from their preferred locations, enhancing convenience and participation rates.	

Data collection method	● To watch user activity, utilise a screen recorder (like OBS Studio). ● SurveyMonkey or Google Forms: To gather satisfaction scores as soon as a task is finished. ● Reasoning: Tools protect user privacy while guaranteeing precise and organised data collecting.
Test material	**Survey for Rating Satisfaction:** Example Question: Throughout the assignment, how pleased were you with Parliament.lk's general usability? (Rate between 1 and 5). Guidelines for Participants: Read the instructions carefully, then do the work in the allocated time. After completing the exercise, note your level of satisfaction right away. Specifics of the Measurement: Task Start: When the participant starts to find the Hansard report.. Task End: When the timer goes off or the participant finds the Hansard report.

4 Study planning and organization

4.1 Experimental Design

INDEPENDENT SAMPLE TEST

In this study, which focuses on satisfaction measurability, the chosen experimental design is a between-subjects design. This design compares the satisfaction levels between two independent age groups: 18-30 years and above 30 years. Since the goal is to assess how satisfied users are with the ease of finding the Hansard report on parliament.lk, the between-subjects design is appropriate for ensuring that each group experiences only one condition.

Rationale for the Design:

The study is designed to evaluate whether there is a significant difference in satisfaction levels between users of different age groups while performing the same task. The task involves locating the Hansard report on parliament.lk.

Null Hypothesis (H_0): "There is no significant difference in satisfaction levels between users of different age groups (18-30 and above 30) in locating the Hansard report on parliament.lk."

Alternative Hypothesis (H_1): "There is a significant difference in satisfaction levels between users of different age groups (18-30 and above 30) in locating the Hansard report on parliament.lk."

The between-subjects design is well-suited for this research because it compares two independent groups, ensuring that no participant is exposed to both age groups. This prevents confounding variables and allows for a direct, unbiased comparison between the groups.

PAIRED SAMPLE T-TEST

This study is concerned with efficiency as the usability metric, particularly assessing the time taken to complete tasks. The experimental design chosen is a within-subjects design, analyzed using a paired-sample t-test. All participants performed the same task finding the Hansard report on parliament.lk twice under two conditions:

- Unguided Workflow: Participants tried the task on their own before starting.
- Guided Workflow: Participants were given clear instructions on how to do the task before making a second attempt.

This design allowed for a direct comparison of task efficiency under both conditions.

Rationale for the Design

The study aims to determine whether providing assistance makes people work better. A within-subjects design paired with a paired-sample t-test is an ideal choice for this due to the following reasons:

1. Direct Comparison of Efficiency: With each participant acting as their control, guidance is easier to interpret in terms of the time taken to complete the task.

2. Reduction of Individual Differences: Differences caused by things like what people already know or their skills are reduced because the same people take part in both situations.

3. The appropriate statistical analysis: A paired-sample t-test will compare the average differences between two related groups, unguided and guided workflows, ensuring the results are accurate and meaningful.

Null Hypothesis (H_0):

"There is no significant difference in task completion times between unguided and guided workflows for participants locating the Hansard report on parliament.lk."

Alternative Hypothesis (H_1):

"There is a significant difference in task completion times between unguided and guided workflows for participants locating the Hansard report on parliament.lk."

Why the Paired-Sample t-Test is Relevant to Efficiency

1. Focus on Efficiency Metrics: The time to complete tasks is in seconds and therefore a continuous variable: a paired-sample t-test would be the appropriate design for analyzing differences in the efficiency metric.

2. Evaluation of Improvements: In the comparison of how the same subjects performed under both conditions, the test finds there were substantial gains in productivity resulting from coaching.
3. Strong Insights: Paired-sample t-test results show how workflow guidance is affecting efficiency, thereby answering the research question on improving.

This experimental design ensures that any change in the duration of tasks is an effect of the new guidance. Within-subjects design coupled with paired-sample t-test really provides a strong and clear way to check efficiency in line with the goals and ideas of this study.

ANOVA

This study uses a within-subjects design as it allows the same participant to test both use cases and records their data. This allows the researchers to ensure that the data is consistent and not affected by external factors. This allows for easier comparison of data

The study aims to evaluate which device type has a more user-friendly interface by comparing the error rate that users experience on each device. To do this, the participant is instructed to complete the task on each device and to record the time taken to complete each of them.

Null Hypothesis (H_0): There is no significant difference in the time taken for the user to retrieve the hansard report between the two device types.

Alternative Hypothesis (H_1): There is a significant difference in the time taken for the user to retrieve the hansard report between the two device types.

The data is then analyzed using the One-Way ANOVA methodology to analyze the differences between each group.

PEARSON CORRELATION

A **within-subjects design** is chosen for this study because it allows the same participants to perform multiple tasks under different conditions. This design ensures consistency in evaluating the usability metric of learnability and minimizes variability caused by individual differences. Here's a more detailed explanation:

- **Consistency Across Participants:** Each participant performs the same set of usability tasks, ensuring that individual differences (such as prior experience or skill level) do not affect the results. This approach provides a controlled comparison, as each participant acts as their own baseline.
- **Efficient Data Collection:** Fewer participants are required since all participants engage in every condition, allowing for richer data collection with less variability.
- **Learnability Focus:** Learnability is best measured by observing how the same individual improves or adapts to tasks over time. A within-subjects design captures these changes effectively.

Study Structure and Data Collection:

- Participants are assigned to one of the two age groups (18-30 years vs. above 30 years) and tasked with locating the Hansard report on the parliament.lk website.
- After completing the task, participants rate their satisfaction using a Likert scale from 1 to 5, where 1 is "Very Dissatisfied" and 5 is "Very Satisfied."
- The between-subjects design ensures a valid comparison of satisfaction scores, where age group serves as the independent variable and the satisfaction score as the dependent variable.
- This experimental design allows for a focused analysis of satisfaction measurability by comparing how different age groups perceive the ease of finding the Hansard report.

4.2 Procedure & Planning

Procedure for Each Session

- **Welcome and Introduction (5 minutes):**
  - Participants were greeted and introduced to the study's purpose..
- **Task Instructions (5 minutes):**
  - The moderator explained that participants would complete two phases of a task involving finding a **Hansard report** on the **parliament.lk** website:
    - **Phase 1:** Locate the Hansard report without any guidance (unguided workflow).
    - **Phase 2:** Receive guidance or step-by-step instructions and attempt the same task again (guided workflow).
  - To ensure consistency, a scripted set of instructions was read to all participants.
- **Task Execution (15-20 minutes):**
  - **Unguided Workflow Task (Phase 1):**
    - Participants were asked to find a Hansard report independently without any assistance or guidance.
    - **Data Collected:** Task completion time for the unguided workflow was recorded, and any observable behaviours, hesitations, or errors were noted.
  - **Guided Workflow Task (Phase 2):**
    - Participants were provided with guidance on how to locate the Hansard report (e.g., step-by-step instructions, tooltips, or a walkthrough).
    - Participants then attempted the task again using this guided workflow.
    - **Data Collected:** Task completion time for the guided workflow was recorded, along with any observable improvements or differences in behaviour.
  - **Given a break for participants (Phase 3):**
    - Participants were given a video to distract them and give a break from the usability study.
  - **Once again asked participants to do the task (Phase 4):**
    - Participants were asked to repeat the task unguided.
    - **Data Collected:** Task completion time for the guided workflow was recorded.
  - **Smartphone Task (Phase 5):**
    - Participants were asked to find a Hansard report independently without any assistance or guidance on a mobile device.
    - **Data Collected:** Task completion time for the guided workflow was recorded.

- **Post-Test Questionnaire (5-10 minutes):**
  - After completing the tasks, participants filled out a Google Form to provide:
    - **Quantitative data:** Task completion times for the unguided and guided workflows.
    - **Qualitative data:** Feedback on their experiences with both workflows, including perceived ease of use and preferences.

Test Duration

- Each session: Approximately 30 minutes, including introduction, task execution, and post-test questionnaire.

- Overall testing period: One week, during which data was collected from 20 participants.

Tasks for Independent-Sample Test Hypothesis

Specific Task:

Participants were first asked to select their age category (18-30 years or above 30 years) and then complete the task of locating the Hansard report on the parliament.lk website. After completing the task, participants were asked to rate their satisfaction with two aspects:

1. **How easy was it to locate the Hansard report on parliament.lk?**
2. **How satisfied are you with the website layout when searching for the Hansard report?**

Why This Activity?
These tasks enable us to collect satisfaction ratings from two independent age groups, 18-30 years and above 30 years, to evaluate whether there is a significant difference in satisfaction levels between the two groups. The participants' ratings, measured on a Likert scale from 1 to 5, serve as the primary data for testing the Independent-Sample Test Hypothesis. By comparing how users from different age groups rate their satisfaction with the ease of finding the Hansard report and the usability of the website layout, we can determine if age influences satisfaction with the website's performance.

Tasks for Paired-Sample Test Hypothesis

- Specific Task:

Participants were then asked to find a Hansard report on the parliament.lk site on two occasions: once with help and once without.

- Why This Activity?

This exercise enables us to compare the time taken to complete tasks with and without guidance. This is the major measurement for testing the hypothesis in a paired-sample test that looks into how guidance affects efficiency.

Tasks for ANOVA Hypothesis

Task 1: Navigate to the Hansard Report on the parliament.lk website for the specific date instructed on laptop or desktop computer and record the time taken to complete.

Task 2: Navigate to the Hansard Report on the parliament.lk website for the specific date instructed on a smartphone and record the time taken to complete.

These two tasks allow us to compare the time taken for the participant to complete the tasks on the two device types and which version of the website has a more user friendly layout.

Task/s for Correlational Hypothesis:

Task 1: Navigate to the Hansard Report on the parliament.lk website for the specific date instructed on laptop or desktop computer and record the time taken to complete.

Task 2: After guiding step by step, do individually to find the Hansard Report from the Parliment.lk website and record the time taken to complete.

These tasks are given to test the Learnability metrics and the pearson correlation test to calculate the correlation coefficient using the time taken to do the task and the user Rating.

4.3 Roles & Responsibilities

In the usability test, several roles were defined to ensure the study ran effectively and met the research objectives. Each role contributed to facilitating the sessions and collecting data efficiently.

1. **Moderator:**
    - The primary role of the moderator was to guide the participants through the usability sessions, ensuring they understood the tasks without providing direct assistance that could influence their performance.
    - Moderators followed a structured script to maintain consistency across all sessions (see *Appendix II: Moderators Information Sheet* for detailed instructions). They provided pre-task instructions, monitored participants during the tasks, and administered post-task surveys or interviews.
    - The moderator was also responsible for ensuring participants remained focused, addressing any technical issues, and recording observations or feedback.
2. **Facilitator:**
    - Facilitators provided logistical support during the sessions, ensuring that the testing environment was set up correctly. They were responsible for preparing the devices, testing the required software, and troubleshooting any technical issues before the sessions began.
    - They also assisted in organizing the schedule, managing participant consent forms, and ensuring adherence to the study timeline.

Moderators instructions:

- Providing a brief introduction to the study and obtaining participant consent.
- Clearly explaining the tasks to participants while avoiding influencing their decision-making or providing guidance.
- Observing and recording participant actions and behaviors without interference.
- Conducting post-task surveys and collecting feedback.

Planning for the Usability Session:

- **Device Setup:** Laptops and smartphones were configured with the necessary software and browser settings to replicate real-world usage conditions.
- **Participant Recruitment:** Participants were recruited in advance and scheduled over multiple days to ensure all sessions could be conducted effectively without time constraints.
- **Session Logistics:** Materials such as consent forms, task instructions, and post-task surveys were prepared in advance. A checklist was also created to ensure every aspect of the session was ready before participants arrived.

4.4 Participant Profile and Recruitment

Participant Profile

This usability study selected participants who were above the age of 18 regardless of prior experience or experience. Such an open approach ensured the study contained different types of user interactions, hence serving useful information on how the application works for a general group of people. That is, it is particularly important to the goals of this study, which want to look at how well the guided and unguided workflows work for all possible users, regardless of background or experience.

Recruitment Methods

Recruitment was done through direct invitations, online messages, and emails sent to people.

- Personal Invitations: People were personally invited to take part in the research study.
- Email Invitations: Invitations were sent via email to individuals and groups to participate in the study. The email included the study's purpose, time duration, and a short description of tasks.

Proof of Hiring Materials

Copies of the recruiting materials, including screenshots from the online postings, sample email messages, and invitation messages, are provided in **Appendix III: Participant Recruitment Evidence.**

They had set up many ways to find participants. That allowed many different types of people to become part of the research.

4.5 Data Collection Methods

Data for this experiment was collected in person, which allowed researchers to interact with the participants directly. At the end of the assigned tasks, participants were invited to submit their responses via a Google Form. The form included questions that would capture both quantitative data and qualitative data.

The test was **unmoderated**, and the moderator's main tasks were to:

- Introduce the tasks and clarify any doubts participants might have about the tasks.
- Make sure the tasks were completed in the most natural manner possible, without interfering with the participants' decision-making process.
- Observe participants' behavior while pointing out any usability issues or challenges faced during the execution of the tasks.

It allows for detailed observation and immediate clarification, which might be necessary, during the in-person setup and moderation. Using the Google Form ensured that responses were systematically recorded and easily accessible for analysis.

4.6 Research Instruments (materials) and Equipment

To collect data for this usability testing session, we used the following tools and materials:

Google Form:

- **Objective**: The form was meant to capture the participant's responses right after finishing the assigned tasks.
- **Content**: The form contained quantitative questions, such as task completion time and Likert scale ratings, and qualitative questions, such as open-ended feedback on the user experience.
- **Justification**: For usability and accessibility, Google Forms has been chosen, which can systematically and efficiently capture the responses. It was also easy to export the data for analysis from this platform.
- **Moderation Tools**: Observational notes were recorded manually by the moderator during the testing session to capture any usability challenges or behavioral cues that would complement the data collected via the Google Form.

These tools have been selected because they are accurate and reliable, while still being user-friendly for the respondents. Google Forms could then make data collection smoother and reduce the need for extra software. Combining this all with direct observation gave a complete set of data that included how tasks were done and user feedback.

For more information about the questionnaire, see **Appendix IV: Data Collection Material.**

5 Report the study results

Independent-sample-test

The results presented in the table below show the satisfaction and ease of locating the Hansard report on parliament.lk for participants in two different age categories: 18-30 and above 30. The research aimed to evaluate whether there is a significant difference in satisfaction and ease of use between these two age groups.

Participant ID	Age Category	Ease of Locating Hansard Report	Satisfaction with Website Layout
P8	18 - 30	3	2
P10	18 - 30	5	5
P11	Above 30	4	5
P16	Above 30	3	3
P26	Above 30	4	3
P27	18 - 30	4	3
P29	18 - 30	5	4
Average	-	4	3.5

For more detail, see **Appendix VI - Raw Data**

Participant Expression / Feeling when doing the task

Participant ID	Expression / Feelings
P8	**Looked a bit confused at the start but seemed okay after finishing.**
P10	**Looked confident and smiled after completing the task quickly.**
P11	**Stayed calm and focused, didn't look frustrated or stuck.**
P16	**Looked a bit annoyed at first but managed to complete it in the end.**
P26	**Didn't show much expression, just focused and got it done.**

Summary of Results:

Ease of Locating Hansard Report: The mean score for the age group 18-30 was 4.0, while the mean for the age group above 30 was 4.0. This indicates that both groups found it equally easy to locate the Hansard report on the website.

Satisfaction with Website Layout: The mean score for the 18-30 age group was 3.57, while the mean for the above 30 group was 3.67. This suggests a slightly higher level of satisfaction with the website layout in the above 30 groups, although the difference is minimal.

Relation to the Research Question

The research question, "How satisfied are you with the ease of finding the Hansard report on parliament.lk?", aimed to assess user satisfaction with the ease of locating the Hansard report on the parliament.lk website across different age categories (18-30 vs. above 30).

The results indicate a slight difference in satisfaction between the two age groups. Users in the above 30 category generally reported slightly higher satisfaction with the ease of finding the Hansard report. This suggests that age may influence the user experience regarding website navigation.This is relevant to the usability metric of satisfaction, as it highlights how the ease of finding the Hansard report may vary based on user age.

Relationship to the Hypothesis

Null Hypothesis (H_0): There is no significant difference in user satisfaction regarding the ease of finding the Hansard report on parliament.lk between the age categories of 18-30 and above 30.

Alternative Hypothesis (H_1): There is a significant difference in user satisfaction regarding the ease of finding the Hansard report on parliament.lk between the age categories of 18-30 and above 30.

The data reveals a slight difference in satisfaction levels between the two age categories. Based on this result, the null hypothesis (H_0) may be challenged, suggesting that there could be a difference in satisfaction based on age.

Paired-sample t-test

Test Results

Participant ID	Time for Unguided Workflow (seconds)	Time for Guided Workflow (seconds)	Difference (Unguided - Guided)
P1	57	06	51
P2	90	20	70
P3	60	30	30
P4	133	17	116
P5	120	09	111
Average	**92**	**16.4**	**75.6**

For more detail, see **Appendix VI - Raw Data**

Explanation of Results

Key Points:

- Unguided Workflow: People took a bit longer to finish the task without guidance, at about 92 seconds on average.(only for 5 participations)
- Guided Workflow: With assistance, participants performed much better and were able to complete the task in an average of 16.4 seconds. (only for 5 participations)
- Time Difference: On average, 75.6 seconds more were spent on unguided workflows than on the guided ones. This indicates that guidance made tasks much more efficient. (only for 5 participations)

Relation to the Research Question

The research question, "**How does task completion time differ when using a guided workflow compared to an unguided workflow?**", aimed to evaluate the efficiency of the guided workflow.

- The results demonstrate that the guided workflow reduces task completion time, confirming that guidance helps users complete tasks more efficiently.
- This supports the usability metric of efficiency, as users performed faster with guided assistance.

Relationship to the Hypothesis

- Null Hypothesis (H_0): There is no significant difference in task completion times between unguided and guided workflows.
- Alternative Hypothesis: H_1 It is far easier to get things done with a guided workflow than an unguided one.

The data indicates that, with support, individuals take less time to complete the tasks. This supports the alternative hypothesis H_1. A paired-sample t-test can be used to statistically determine if this difference is significant.

ANOVA

Test Results

Participant	Desktop	Mobile
P1	5	7
P2	50	50
P3	70	70
P4	14	26
P5	60	60
P6	60	60
P7	56	60
P8	30	83
P9	70	70
P10	50	63
P11	8	14
P12	120	120
P13	69	65
P14	240	240
P15	80	70
P16	22	50
P17	15	19
P18	60	60
P19	8	8
P20	15	20
P21	40	60
P22	65	83
P23	71	90
P24	10	10
P25	50	40
P26	70	55
P27	45	80
P28	55	50
P29	50	111

SUMMARY				
Groups	*Count*	*Sum*	*Average*	*Variance*
Desktop	28	1508	53.85714286	2084.42328
Mobile	28	1683	60.10714286	1999.358466

ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	546.875	1	546.875	0.2678277313	0.6069063273	4.019540764
Within Groups	110262.1071	54	2041.890873			
Total	110808.9821	55				

Twenty-Nine participants completed a navigation task to locate the hansard report for a given date on the parliament.lk website on two different device types (Laptop/Desktop and Smartphone).

The One-Way ANOVA test using task completion time as independent variable and group as dependent variable suggests that there is not a significant difference between the two conditions.

Therefore the Null hypothesis is confirmed.

Pearson correlation r-test

Test Results

Participants ID	Final Task Time (in Seconds)	Confidence Ratings(1-5 Likert scale)
P6	**60**	**4**
P8	**30**	**1**
P12	**120**	**3**
P15	**80**	**4**
P19	**08**	**5**
P21	**40**	**3**
P23	**71**	**5**
P24	**10**	**4**
P26	**70**	**4**
Average	**54.33**	**3.66**

For more detail, see **Appendix VI - Raw Data**

Explanation of the results

Key Insights

- **Correlation Value:** A weakly positive link is shown by the correlation coefficient (r = 0.2355). This indicates that there may be a minor rise in confidence ratings as the end task duration increases, although the link is weak.
- Usability Implication: A poor association suggests that user confidence ratings may be more significantly influenced by variables other than task time.

Relationship to the Research Question

- The research question examines the relationship between usability factors (final task time) and the effectiveness of first-time users (confidence ratings) in navigating the Parliament.lk website.
- The weak positive correlation suggests that task time is not a major determinant of user confidence.

Research to the Hypothesis

- **Null Hypothesis (H_0):** There is no significant correlation between usability factors and the effectiveness of first-time users in learning to navigate and perform essential tasks on the website.
- **Alternative Hypothesis (H_1):** There is a significant correlation between usability factors and the effectiveness of first-time users in learning to navigate and perform essential tasks on the website.

There is insufficient evidence to reject the null hypothesis because of the poor connection between task completion time and confidence, as indicated by the computed correlation of 0.2355. The alternative hypothesis, according to which usability considerations have a major impact on first-time visitors' ability to navigate the site, is not supported by the correlation.

Conclusion:

Although task completion time and confidence ratings have a small positive association, it is possible that other factors—aside from task timing—have a greater influence on how well new users can utilize the website and complete activities.

6 Ethics form

For more detail, see **Appendix VII: Ethics Form**. Attached is a copy of the ethics application form and attachments submitted.

7 Conclusions

This research did not make use of a pilot test. Due to this there were a few issues faced when collecting the data that could have been caught early otherwise. One such issue was the difficulty in getting accurate timing due to participant error. To rectify this we could use a moderated survey instead so that participants can get clarifications as they go through.

Appendix

Appendix I. Participant Information Sheet

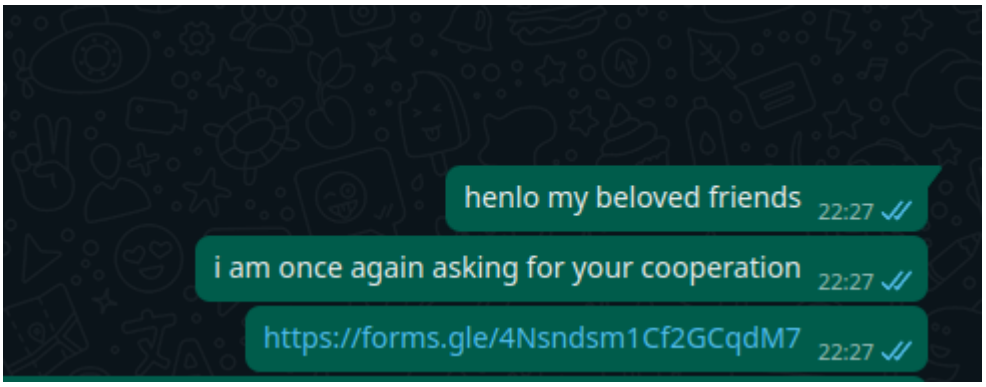
https://docs.google.com/document/d/1FMHtoadxWKCbwxl_mR2hYCticgBrep59/edit?usp=drive_link&ouid=102598590118639191071&rtpof=true&sd=true

Appendix II. Moderators Information Sheet

https://docs.google.com/spreadsheets/d/1ooDv6RHfszjLVF7jN6TiuP6drSs6TWmISVYU8npC3NY/edit?usp=sharing

Appendix III. Participant Recruitment Evidence

Below is a screenshot of a personal invitation to members of the target demographic requesting them to participate in the survey.



Appendix IV. Data Collection Material

https://docs.google.com/forms/d/1IHmn1XJAFKM6t5KgHiOx1XPYYie79Qh9sm7jX4ocTkY/edit

Appendix V. Test Recordings

https://drive.google.com/file/d/1SPTdOBj0jHisgnD0m4TaXyyf4LbDOSFU/view?usp=drive_link

Appendix VI. Raw data

https://docs.google.com/spreadsheets/d/1RaPi9YuBrfaauKYLykXNKOpvGT0TwOpVVy9SwvVkRCk/edit?usp=sharing

Independent-sample-test Raw data

Participant ID	Age Category	Ease of Locating Hansard Report	Satisfaction with Website Layout
P1	18-30	3	3
P2	18-30	3	5
P3	18-30	2	5
P4	18-30	3	4
P5	18-30	4	3
P6	18-30	3	3
P7	18-30	3	2
P8	18-30	2	1
P9	18-30	5	5
P10	Above 30	4	5
P11	18-30	3	2
P12	18-30	3	3
P13	18-30	5	5
P14	18-30	4	4
P15	Above 30	3	3
P16	18-30	3	2
P17	18-30	4	3
P18	18-30	2	2
P19	18-30	5	3
P20	18-30	4	3
P21	18-30	3	3
P22	Above 30	2	3
P23	Above 30	3	2
P24	18-30	3	3
P25	Above 30	4	3
P26	18-30	4	3
P27	18-30	3	2
P28	18-30	5	4
P29	Above 30	2	2

https://docs.google.com/spreadsheets/d/1RaPi9YuBrfaauKYLykXNKOpvGT0TwOpVVy9SwvVkRCk/edit?usp=sharing

Paired-sample t-test - Raw data

Participant ID	Time for Unguided Workflow	Time for Guided Workflow
P1	0:00:57	0:00:06
P2	0:01:30	0:00:20
P3	0:01:00	0:00:30
P4	0:02:13	0:00:17
P5	0:02:00	0:00:09
P6	0:01:20	0:00:24
P7	0:05:00	0:00:33
P8	0:01:45	0:00:07
P9	0:01:10	0:00:06
P10	0:01:31	0:00:28
P11	0:00:54	0:00:21
P12	0:10:00	0:00:17
P13	0:00:10	0:00:09
P14	0:04:00	0:00:25
P15	0:01:10	0:00:30
P16	0:03:12	0:00:07
P17	0:00:32	0:00:06
P18	0:02:00	0:00:26
P19	0:01:26	0:00:33
P20	0:00:05	

https://docs.google.com/spreadsheets/d/1RaPi9YuBrfaauKYLykXNKOpvGT0TwOpVVy9SwvVkRCk/edit?usp=sharing

Pearson Correlation RAW Data

https://docs.google.com/spreadsheets/d/1RaPi9YuBrfaauKYLykXNKOpvGT0TwOpVVy9SwvVkRCk/edit?usp=sharing

ANOVA RAW Data

Participant	Desktop	Mobile
P1	0:00:05	0:00:07
P2	0:00:50	0:00:50
P3	0:01:10	0:01:10
P4	0:00:14	0:00:26
P5	0:01:00	0:01:00
P6	0:01:00	0:01:00
P7	0:00:56	0:01:00
P8	0:00:30	0:01:23
P9	0:01:10	0:01:10
P10	0:00:50	0:01:03
P11	0:00:08	0:00:14
P12	0:02:00	0:02:00
P13	0:01:09	0:01:05
P14	0:04:00	0:04:00
P15	0:01:20	0:01:10
P16	0:00:22	0:00:50
P17	0:00:15	0:00:19
P18	0:01:00	0:01:00
P19	0:00:08	0:00:08
P20	0:00:15	0:00:20
P21	0:00:40	0:01:00
P22	0:01:05	0:01:23
P23	0:01:11	0:01:30
P24	0:00:10	0:00:10
P25	0:00:50	0:00:40
P26	0:01:10	0:00:55
P27	0:00:45	0:01:20
P28	0:00:55	0:00:50
P29	0:00:50	0:01:51

https://docs.google.com/spreadsheets/d/1RaPi9YuBrfaauKYLykXNKOpvGT0TwOpVVy9SwvVkRCk/edit?usp=sharing

Appendix VII. Ethics form

Cover Sheet:
https://drive.google.com/file/d/1U0jNUSXpelMj1M8FKBQf-QTCD4F2Fe-j/view?usp=drive_link

Part A:

https://drive.google.com/file/d/1-OpgmolQB2kqg3lPvvMR1DD97W6oCG2t/view?usp=drive_link